

# SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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| Course: 21CSC203P — Advanced Programming Practice | Department: Computer Science and Engineering |

## Unit II | Session 3 | SLO 1: Inheritance & Method Overriding (Code: 2031)

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**Q: Activity: Implement Single and Multilevel Inheritance representing an Employee hierarchy. Create a superclass Employee and subclasses Manager and Executive. Demonstrate super constructor invocation and method overriding.**

**Ans: Java Implementation:**

```
class Employee {
    String name;
    double basicSalary;

    Employee(String name, double basicSalary) {
        this.name = name;
        this.basicSalary = basicSalary;
    }

    double calculateSalary() {
        return basicSalary;
    }
}

class Manager extends Employee {
    double bonus;

    Manager(String name, double basicSalary, double bonus) {
        super(name, basicSalary); // Super constructor
        this.bonus = bonus;
    }

    @Override
    double calculateSalary() {
        return super.calculateSalary() + bonus;
    }
}

public class InheritanceDemo {
    public static void main(String[] args) {
        Employee emp = new Employee("Alice", 50000);
        Manager mgr = new Manager("Bob", 70000, 15000);
        System.out.println("Employee Salary: " + emp.calculateSalary());
        System.out.println("Manager Salary: " + mgr.calculateSalary());
    }
}
```